- Precision
- Random numbers
- Language extensions

Number Systems

A number system is simply a set of symbols (digits) used to represent numbers. For example, the decimal system uses ten digits; they are 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. These should be instantly familiar; it's been quite a popular system for representing numbers from as far back as the 1st century. As we count our way through the cardinal numbers (positive integers, starting at one), we eventually run out of digits (in the decimal system this happens when we reach ten), at which point we begin to use two digits (when we reach one hundred), then three (when we reach one thousand), and so on.

The number of digits used in a number system is called the *base* (or more formally the *radix*). Decimal is Base 10, and feels natural to humans because, well, we (typically) have ten convenient counters on our hands. Binary is Base 2 (using the digits 0 and 1), which is more natural for electromechanical or electronic systems built from arrays of switches (also known as computers). Some cultures use other bases such as six (Southern New Guinea), twenty (Aztec), or even sixty (Sumerian, Babylonian - we still use Base 60 numbering today for measuring time, trigonometry and other things).

PHP provides built-in functionality for four common number systems for integer (whole number) operations: binary (Base 2), octal (Base 8), decimal (Base 10), and hexadecimal (Base 16). In case you haven't encountered a number system with a base larger than ten before, when we've exhausted the digits 1–9, we represent the remaining numbers with letters (A-F in hexadecimal). So ten in hex is A, fifteen is F, sixteen is 10, twenty-six is 1A and so on. There is also support in PHP for arbitrary number bases (up to Base 36), the notation for this base combines digits 0-9 and the letters A-Z.

Non-decimal Integer Literals

To demonstrate how to represent the same number using PHP's four different systems, let's first pick an interesting number: the Ramanujan-Hardy number *1729* will do nicely. (If you don't know the story, the mathematician Srinivasa Ramanujan realized some interesting properties of this number while seriously ill in hospital. His friend and